

Ankush Pratham

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GitHub | LinkedIn | Portfolio

Summary

Computer Science undergraduate at VIT Chennai with a strong foundation in Data Structures & Algorithms, backend engineering, full-stack development, and applied machine learning. Experienced in building REST APIs, distributed systems, data pipelines, and database-backed applications using Python, FastAPI, Next.js, PostgreSQL, Redis, and MongoDB. Seeking a Software Engineering or Backend Engineering Internship.

Technical Skills

Languages: C/C++, Python, Java, JavaScript, R
Software Engineering: Data Structures & Algorithms, OOP, System Design, REST APIs, API Design
Frameworks: FastAPI, Next.js, React, Streamlit, Tailwind CSS
Databases & Infrastructure: PostgreSQL, MySQL, MongoDB, Redis, Docker
Data & ML: Pandas, NumPy, Scikit-learn, TensorFlow, Matplotlib, EDA, NLP, RAG Pipelines
Tools & Cloud: Git, GitHub, GitHub Actions, AWS, Vercel, Linux, Jupyter Notebook
Testing: pytest, JUnit, Postman

Education

Vellore Institute of Technology, Chennai Expected 2028
B.Tech in Computer Science Engineering CGPA: 8.38/10

Experience

Grus & Grade Pvt. Ltd. | Delhi, India 2 Months
Software Engineering Intern

- Reduced manual data-cleaning time by 40% by engineering end-to-end Python data-processing pipelines for industrial datasets in an agile development environment.
- Improved reliability of internal analytics dashboards by developing REST APIs, writing pytest-based tests for API endpoints, and participating in peer code reviews.

Projects

Distributed FastAPI Rate Limiter *Python, FastAPI, Redis, Docker, pytest*

- Built distributed rate-limiting middleware using Redis as shared state, enforcing per-client request quotas across multiple service instances.
- Implemented Sliding Window and Token Bucket algorithms with TTL-based resets and pytest integration tests for quota enforcement and HTTP 429 responses.

Focus.gov – AI-Powered UPSC Preparation Platform *Next.js, PostgreSQL, REST APIs, AI APIs*

- Built and deployed a full-stack AI learning platform with REST APIs, PostgreSQL, authentication, and LLM-powered content generation serving 500+ users.
- Developed personalized study workflows and AI-generated learning content, reducing estimated manual content-creation time by 50%.

RAG Data Loss Analyzer *Python, Streamlit, Pandas, MongoDB, LLM*

- Built a platform that analyzes CSV datasets, identifies missing data, and generates business-impact summaries using an LLM-powered RAG pipeline.
- Engineered vector retrieval and contextual generation to produce data-grounded insights with persistent storage and secure session management.

News Category Classifier *Python, Scikit-learn, Pandas, NLP*

- Built a multi-class NLP classifier for 1,000+ news articles across 5 categories, achieving 93% accuracy using TF-IDF and text preprocessing.
- Compared Naive Bayes and Logistic Regression using precision, recall, and F1-score, improving F1-score by 5% over baseline.

Certifications

- LaunchED — Data Science & AI/ML (2025)
- Complete Data Science, Machine Learning, Deep Learning & NLP Bootcamp (2026)
- Mathematics: Basics to Advanced for Data Science and GenAI (2026)